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DYNAMIC CORRECTION OF THE STEERING-CHARACTERISTIC CURVE AND APPLICATION TO AN EPS CONTROL SYSTEM

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ABSTRACT–The steering torque of an electric-power steering (EPS) motor is interrelated with the performance of the EPS control system, therefore calculating an exact steering torque is critical in this application. This study presents a dynamic correction method that greatly decreases the calculated error in the steering torque; the PID controller demonstrated here is therefore suitable for the demands of this system. Based on an analysis of the detection process of the steering torque sensor, we first deduced that the variation of the system resistance torque equals the difference between the measured value of the steering torque and the ideal one in the previous cycle. Based on this result, we then proposed a dynamic correction method. Finally, a comparison of the simulated and experimental results for several vehicles evidenced the effectiveness of this dynamic correction method.

KEY WORDS : EPS, Steering characteristic, PID control, Road feel

NOMENCLATURE

$\hat{T}_m$: reference motor torque [$N{\cdot}m$]
T_m : actual assisting torque [$N{\cdot}m$]
T_d : actual steering torque [$N{\cdot}m$]
T_r : system resistance [$N{\cdot}m$]
$\hat{T}_d$: ideal steering torque [$N{\cdot}m$]
ΔT_r : change of system resistance [$N{\cdot}m$]
θ_s : angle of the steering wheel [rad]
J_s : rotational inertia of the steering wheel [$kg{\cdot}m^2$]
K_s : stiffness of the steering wheel [$N{\cdot}m/rad$]
C_s : damping of the steering wheel [$N{\cdot}m{\cdot}s/rad$]
θ_c : rotational angle of the export shaft [rad]
J_c : rotational inertia of the export shaft [$kg{\cdot}m^2$]
C_c : damping of the export shaft [$N{\cdot}m{\cdot}s/rad$]
θ_m : rotational angle of the motor [rad]
J_m : rotational inertia of the motor [$kg{\cdot}m^2$]
K_m : stiffness of the motor [$N{\cdot}m/rad$]
C_m : damping of the motor [$N{\cdot}m{\cdot}s/rad$]
u : armature voltage of the motor [V]
k_α : torque constant of the motor [$N{\cdot}m/A$]
k_b : reverse electromotive force coefficient of the motor [$V{\cdot}s/rad$]
R : armature resistance of the motor [Ω]
G : steering gear radio
T_c : equivalent system resistance of the export shaft [$N{\cdot}m$]
T_{sen} : measuring torque of the steering torque sensor [$N{\cdot}m$]
P_m : EPS motor nominal power [W]
$I_{\max}$: EPS maximum current [A]

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1. INTRODUCTION

Electric power-steering (EPS) systems are all-electric, engine-independent systems using an electric motor to provide power-steering assistance (Lin and Shi, 2001); they have many advantages over traditional hydraulic power-steering systems. An EPS eliminates the need for a power-steering pump, hydraulic fluid, hoses, a drive belt and a pulley on the engine; therefore, the total system is lighter than a traditional one and the corresponding maintenance is easier. Additionally, as the EPS system is an on-demand system that operates only when the steering wheel is turned, the fuel efficiency of a vehicle equipped with such system is up to 3% better than that of vehicles equipped with an equivalent-output hydraulic system (Nakayama and Suda, 1994).

The typical EPS system consists of four parts: steering mechanism, torque sensor, steering motor and controller. When the torque sensor detects the driver's steering torque, the controller communicates the target torque setting unit to compute $\hat{T}_m$, which is interrelated with the performance of the EPS system (Lin and Shi, 2001; Nakayama and Suda, 1994; Nozaki, 2006). It is easy to see that calculating an exact steering torque is very important in an EPS control system.

Usually, the controller calculates $\hat{T}_m$ using the steering-characteristic curve and T_d (Liu, 2007; Li and Xu, 2004).

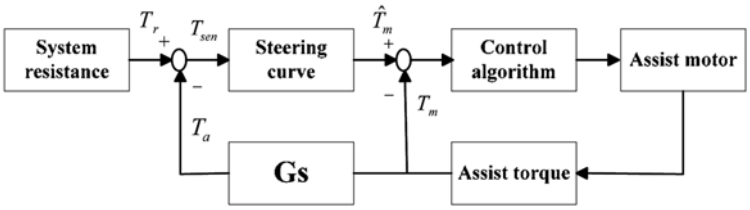


Figure 1. Block diagram of the EPS control system.

The steering-characteristic curve is static and does not change with the current T_m, but the control signals of the torque-assisting motor are discrete, and thus, the motor-assisted torque does not vary with time. Therefore, if $\hat{T}_m$ is calculated with this T_d value, it causes an error in $\hat{T}_m$, and it is difficult to eliminate this error for the latter PID control algorithm, even though T_m is the negative feedback.

In recent studies, several advanced control algorithms such as predictive control (Liu, 2007), fuzzy control (Li and Xu, 2004) and optimal control (Wu *et al*., 2004) have been employed in EPS control systems with the aim of eliminating the calculational error. However, few studies have focused on correcting the static curve dynamically to eliminate the reference-torque error and so simplify the design of the PID controller required to meet the system demands.

The purpose of this paper is to propose a dynamic correction method for the steering-characteristic curve, which can greatly decrease the calculational error of the steering torque and allow the PID controller used in this paper to meet the demands of an EPS.

2. DOUBLE-LOOP, NEGATIVE-FEEDBACK CONTROL OF AN EPS SYSTEM

As shown in Figure 1, the EPS control system is a double-loop feedback system, and the two feedback variables are both the actual assisting torque of the motor T_m. In the inside feedback-control loop, the control of the difference between T_m and $\hat{T}_m$ is used to ensure that T_m follows $\hat{T}_m$ rapidly and effectively. In the outside feedback-control loop, the steering curve is designed to calculate $\hat{T}_m$ based on the vehicle speed and the sensor torque, where the sensor torque equals the difference between T_r and T_m. Thus, it is easy to see that the controllers in the two feedback loops supplement for each other: a good steering curve in the outside loop can be used to calculate an exact $\hat{T}_m$ for the inside control, which makes the following $\hat{T}_m$ more accurate; additionally, a fast and efficient following $\hat{T}_m$ in the inside feedback can also eliminate the calculated error in $\hat{T}_m$.

Therefore, there are two issues to be addressed. The first is to design an improved steering curve to calculate $\hat{T}_m$ and the second is to select a proper control algorithm to make T_m follow $\hat{T}_m$ more quickly and accurately.

Regarding the first issue, the PID control algorithm presented herein is a pioneering one. Compared with other advanced algorithms, the advantages such as a simpler structure, a shorter design cycle, a lower cost and easier executions are obvious. It is known that compensating for the errors due to the inherent characteristics of the control system, such as sensor delay and executive motor delay, can be effective; however, calculating $\hat{T}_m$ without considering the influence of the frequency and the variation in the value of T_r will cause an error in $\hat{T}_m$, so it is difficult to make a simple PID controller meet the demands of the system.

Regarding the second issue, there are three typical steering-characteristic curves used in the design of EPS control systems: linear, bilinear and curvilinear steering characteristic curves. In Kim and Song (2002) and Liao and Du (2003), the authors discussed the steering curves and concluded that the curvilinear form, in particular, the "bath-tub curve", was the best. With the increase of the steering torque T_d, the slope of this curve should increase gradually from zero and then decrease gradually to zero; this curve exactly reproduces the conventional hydraulic steering curve.

In the next section, a dynamic correction method for the steering-characteristic curve is proposed, which allows the PID controller to meet EPS control demands.

3. DYNAMIC CORRECTION OF THE STEERING CURVE

3.1. Ideal Steering Torque $\hat{T}_d$ and Actual Steering Torque T_d

To minimize the calculational error in $\hat{T}_m$ based on the static steering curve, the relation between $\hat{T}_d$ and T_d should be discussed. At a given T_r, a steering torque can be back-calculated from the steering-characteristic curve. We define this torque as the ideal one, $\hat{T}_d$, and the actual measuring torque is expressed by T_d; the relationship of $\hat{T}_d$ and T_d is shown in Figure 2.

Among the delays of the EPS control system, the delay of the torque sensor is larger than the delay of the DC motor. Compared with the cycle of the algorithm, the response time of the motor is short. For simplicity, the response time of the motor is ignored here, yielding:

$$T_m = \hat{T}_m = f(\hat{T}_d) \tag{1}$$

At the time point, the system resistance is expressed as T_r

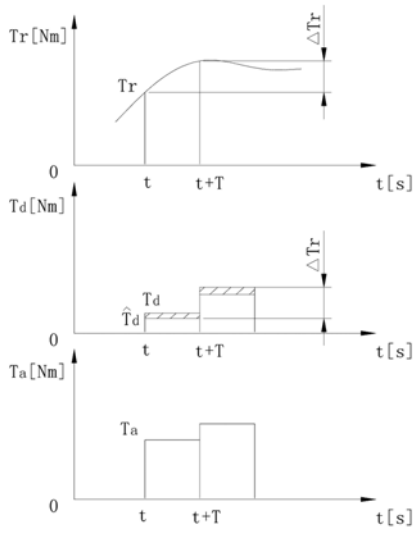


Figure 2. Relational diagram between $\hat{T}_d$ and T_d.

(t) and, according to the definition of $\hat{T}_d$:

$$T_r(t) = T_m(t) + \hat{T}_d(t) \tag{2}$$

If the time point before next cycle ends is defined as $t+T-0$, then T_r and T_m at this time and T_d at the time $t+T$ will be the balance torques:

$$T_r = (t+T-0) = T_m(t+T) + T_d(t+T) \tag{3}$$

and during the time $t+0$ to $t+T-0$, the motor-assisting torque T_m does not vary and T_r is a successive variable.

Thus,

$$T_m(t+T-0) = T_m(t) \tag{4}$$

$$T_r(t+T-0) = T_r(t+T) \tag{5}$$

Substituting Equations (4) and (5) into Equation (3):

$$T_r(t+T) = T_m(t) + T_d(t+T) \tag{6}$$

then combining Equations (2) and (6) gives:

$$T_r(t+T) - T_r(t) = T_d(t+T) - \hat{T}_d(t) \tag{7}$$

For simplicity, the change of system resistance ΔT_r is defined as follows:

$$\Delta T_r = T_r(t+T) - T_r(t) \tag{8}$$

Then, Equation (7) can be expressed as:

$$\Delta T_r = T_d(t+T) - \hat{T}_d(t) \tag{9}$$

Thus, the variation of the system-resistance torque ΔT_r at the time $t+T$ equals the difference between the measured steering torque $T_d(t+T)$ and the ideal one in the previous cycle $\hat{T}_d(t)$.

3.2. Calculation of the Reference Assisting Torque $\hat{T}_m$

As discussed above, the change in the system resistance ΔT_r is related to the measured torque $T_d(t+T)$ and $\hat{T}_d(t)$ in this computing cycle; thus, the function $\hat{T}_m(t+T) = f(\hat{T}_m(t+T))$ can be expressed $\hat{T}_d(t)$ as and ΔT_r, deduced as follows:

$$\hat{T}_m = f(\hat{T}_d) \tag{10}$$

$$T_r = \hat{T}_m + \hat{T}_d \tag{11}$$

Combining Equations (10) and (11):

$$\hat{T}_m = f(T_r - \hat{T}_m) \tag{12}$$

This is an implicit function of $\hat{T}_m$, and we can therefore assume that $\hat{T}_m = g(T_r)$ because $\hat{T}_m$ can be shown to be monotonically increasing with respect to T_r. Using Taylor's Formula for $\hat{T}_m(t+T)$ at the point $T_r(t)$, and neglecting the higher-order terms because ΔT_r is far less than 1 Nm in an algorithm cycle of 1 ms, we have:

$$T_m(t+T) \doteq g(T_r(t)) + \frac{dg(T_r(t))}{dT_r(t)} \cdot [T_r(t+T) - T_r(t)] + \frac{d^2 g(T_r(t))}{dT_r^2(t)} \cdot \frac{[T_r(t+T) - T_r(t)]^2}{2} \tag{13}$$

Substituting Equation (8) into Equation (13) and rearranging the calculating formula for the ideal reference assisting torque, we obtain:

$$\hat{T}_m(t+T) \doteq A + B \cdot \Delta T_r + C \cdot \frac{\Delta T_r^2}{2} \tag{14}$$

where: $A = g(T_r(t)) = \hat{T}_m(t)$

$$B = \frac{dg(T_r(t))}{dT_r(t)} = \frac{\dot{f}(T_d(t))}{1 + G_s \cdot \dot{f}(\hat{T}_d(t))}$$

$$C = \frac{d^2 g(T_r(t))}{dT_r^2(t)} = \frac{d\left(\frac{dg(T_r(t))}{dT_r(t)}\right)}{dT_r(t)} = \frac{d\frac{dg(T_r(t))}{dT_r(t)}}{d\hat{T}_d(t)} \cdot \frac{d\hat{T}_d(t)}{dT_r(t)} \quad ;$$

$$= \frac{\ddot{f}(\hat{T}_d(t))}{[1 + G_s \cdot \dot{f}(\hat{T}_d(t))]^2} \cdot \frac{1}{1 + G_s \cdot \dot{f}(\hat{T}_d(t))}$$

$$= \frac{\ddot{f}(\hat{T}_d(t))}{[1 + G_s \cdot \dot{f}(\hat{T}_d(t))]^3}$$

Thus, the correcting coefficients (A, B and C) are functions of the ideal steering torque $\hat{T}_d(t)$, and this torque can be back-calculated from the computed result $T_m(t)$ in time t and the steering-characteristic curve. Thus, using the steering-characteristic curve, we can calculate $T_m(t+T)$,

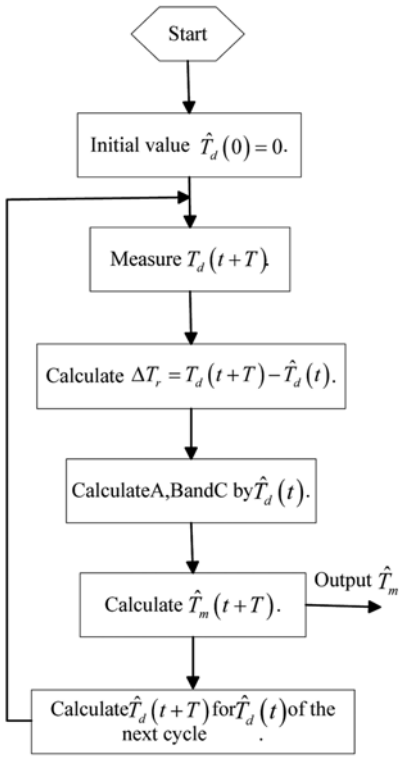


Figure 3. Flowchart of calculating the reference toque $\hat{T}_m$.

from $\hat{T}_d(t)$, $T_d(t+d)$ and v. This calculation technique is based on the variation of the system loads; the detailed process is shown in Figure 3.

It is apparent that the correcting coefficients (A, B and C) form the core of the dynamic correction for the steering curve, and they are variable parameters. Combining these with the latter PID control algorithm, the whole control consists of the PID control algorithm with variable parameters. The difference is that this control algorithm has been divided into two parts: the variable-parameter part is the dynamic correction of the steering curve and the former PID control is the fixed-parameter part.

Simultaneously, we know that to correct the steering curve dynamically, some complicated equations are required to calculate the correcting coefficients (A, B and C). Fortunately, if a static steering curve is given, the correcting coefficients can first be calculated from an array of $\hat{T}_d$ and then a lookup table and interpolation can be utilized to calculate these coefficients to obtain the new T_m, greatly facilitating the calculation process.

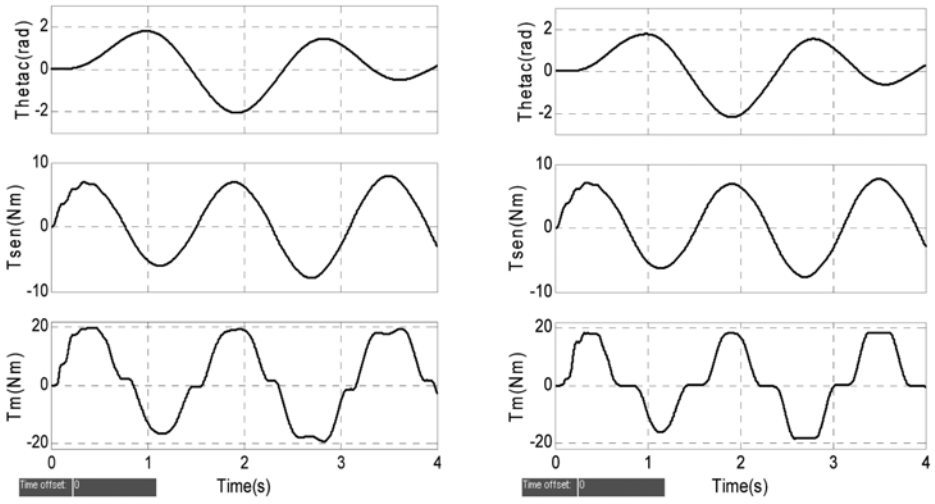
We note that this applies only to the calculation of $\hat{T}_m$ in an EPS control system, while other issues such as the process of selecting PID parameters, damping control, and return-to-center control were not addressed here because they are not the main focus of this work.

4. SIMULATION OF THE DYNAMIC CORRECTING PERFORMANCE FOR THE ASSISTING CHARACTERISTIC

In the following paragraphs, we chose to analyze the characteristics of a familiar steering-column-type EPS system as a simulation example. Here, three centralized masses (the steering wheel, the motor rotor and the export-steering shaft) were considered in the simulation, for which the kinematic equations are given by (H. Nozaki, 2006; Wu *et al*., 2007; Jin *et al*., 2007; Gao *et al*., 2007; Zhang *et al*., 2008) :

$$J_s\ddot{\theta}_s = T_d - K_s(\theta_s-\theta_c) - C_s\dot{\theta}_s \tag{15}$$

$$J_c\ddot{\theta}_c = K_s(\theta_s-\theta_c) + GK_m(\theta_m - G\theta_c) - T_c(\theta_c,\dot{\theta}_c) \tag{16}$$



(a) Calculating T_m through the static steering curve (b) Calculating T_m through the dynamic correction method

Figure 4. Comparison of the response curves of θ_c, T_{sen} and T_m to driver's sinusoidal input torque at frequency of 4 rad/s and an amplitude of 7 Nm.

Table 1. Simulation parameters.

Parameter symbol	Units	Value
J_s	$kg{\cdot}m^2$	0.0459
K_s	$N{\cdot}m/rad$	114.6
C_s	$N{\cdot}m{\cdot}s/rad$	0.361
J_c	$kg{\cdot}m^2$	32
C_c	$N{\cdot}m{\cdot}s/rad$	0.023
J_m	$kg{\cdot}m^2$	0.000471
K_m	$N{\cdot}m/rad$	3.02
C_m	$N{\cdot}m{\cdot}s/rad$	0.003399
k_α	$N{\cdot}m/A$	0.02
k_b	$V{\cdot}s/rad$	0.05
R	Ω	0.1
G		16.5:1

$$J_m\ddot{\theta}_m = \frac{k_\alpha}{R}(u - k_b\dot{\theta}_m) - K_m(\theta_m - G\theta_c) - C_m\dot{\theta}_m \tag{17}$$

The simulation is conducted in MATLAB7.0 /Simulink, and the setting parameters are summarized in Table 1. T_d is the simulation input, and the simulation outputs are θ_c, T_{sen} and T_m, where $T_{sen} = K_s(\theta_s - \theta_c)$ and $T_m = K_m(\theta_m - G\theta_c)$.

Figure 4 shows a comparison of the responses of θ_c, T_{sen} and T_m to a sinusoidal torque input at a frequency of 4 rad/s and an amplitude of 7 Nm. Due to the smooth variation of the steering torque, there were no differences between the responses of θ_c and T_{sen} whether or not the dynamic correction method was used; however, when the assistance of the motor reached the saturation state, the T_m response without the dynamic correction method had a larger overshoot and lag than required, and T_m could not return to zero when the steering wheel returned to the center. In contrast, using the dynamic correction method, a better T_m response was obtained, no overshoot or lag was observed, and this T_m response returned to zero.

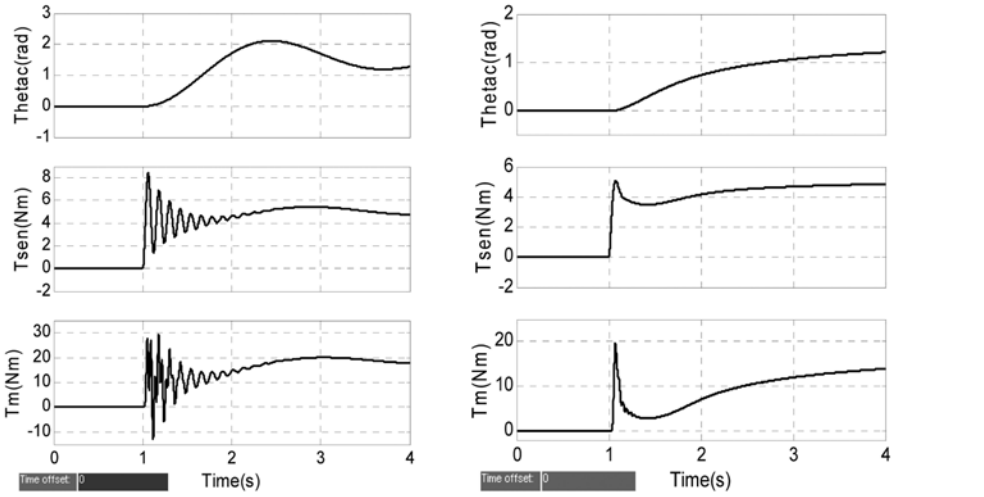
Figure 5 shows the compared responses of θ_c, T_{sen} and T_m to a step torque input at a given final value of 5 Nm. It is evident that the outputs of T_{sen} and T_m had larger overshoots and more high-frequency ripples than required and that a larger amplitude exists in the output of θ_c, causing an obvious sway in the steering wheel. In contrast, using the dynamic correction method, the tracking performance was remarkably improved, the high-frequency ripples of T_{sen} were nearly eliminated, the response of θ_c became smoother and no overshoot exists, which is important for the driver's comfort; the overshoot and high-frequency ripple of T_m were also greatly reduced. Together, these prove the effectiveness of the dynamic correction method presented in this paper.

5. EXPERIMENTS

To investigate the effects of changing the calculating



Figure 6. Experimental vehicle and NI instruments.



(a) Calculating T_m from the static steering curve (b) Calculating T_m with the dynamic correction method

Figure 5. Comparison of the response curves of θ_c, T_{sen} and T_m to a step torque input (0–5 Nm).

Table 2. Vehicle and PID parameters.

Parameters	Units	Value
m	kg	1,100
r	m	5.25
u	V	12
P_m	W	170
I_{max}	A	30
G		16.5:1

formula for the reference torque, experiments with and without dynamic correction were conducted using an China vehicle, as shown in Figure 6. Using USA National Instruments hardware and the associated LabView software, the resulting data were collected and displayed. In the experiment, the selected lemniscate tests were conducted at the speed of 20 km/h on a concrete road, and the EPS control system was controlled by the simple PID algorithm. The test vehicle and PID parameters are listed in Table 2.

As shown in Figure 7, when calculating $\hat{T}_m$ directly through T_d the system displayed some obvious ripples or vibrations in the experimental steering curve, reaching maxima when the steering was initiated and the wheel returned equal to 4.97% and 4.32%, respectively, which are much greater than the permitted value of 2%, above which drivers experience poor handling. Under the same conditions, calculating $\hat{T}_m$ with the dynamic correction method nearly completely eliminated these problems; the steering and returning vibrations were reduced to 1.43% and 1.87%, demonstrating the quality of the simulation results.

6. CONCLUSIONS

In this study, we analyzed the detection process of the steering-torque sensor, distinguished the ideal steering torque from the actual one, and then deduced that ΔT_r at time $r+T$ equals the difference between the measuring value of the steering torque $T_d(t+T)$ and the ideal one $\hat{T}_d(t)$ of the previous cycle.

Using this method, we proposed a dynamic correction method for the steering-characteristic curve that greatly decreased the calculated error in the reference torque and demonstrated that the PID controller used herein met the system demands.

Finally, the simulated and experimental results on real vehicles showed the effectiveness of the dynamic correction method, as this method effectively solved the vibration problem in the assisting torque when increasing or decreasing the steering torque abruptly.

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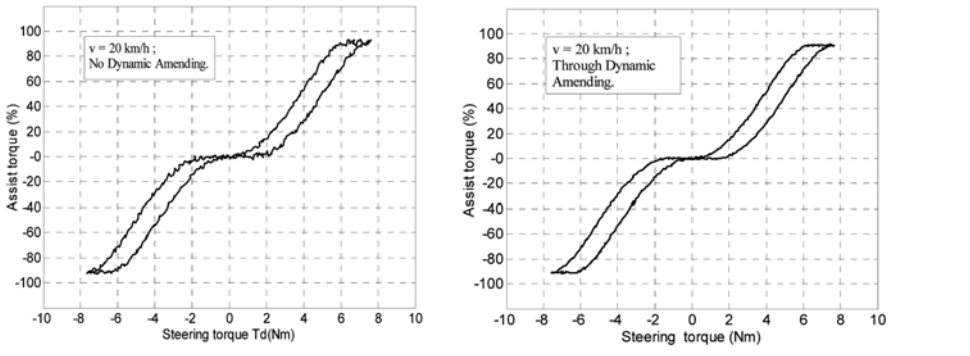
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(a) Calculating $\hat{T}_m$ through the static steering curve (b) Calculating $\hat{T}_m$ through the dynamic correction method

Figure 7. Experimental steering curves with and without dynamic correction.

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